CONGRUENT PLANE PROJECTIONS DO NOT DETERMINE A CONVEX BODY UP TO TRANSLATION OR CENTRAL INVERSION

ARGUS OPEN RESEARCH PROJECT

Abstract. We give an explicit, analytically and rationally certified reconstruction of a pair of compact full-dimensional C^2 convex bodies $K, L \subset \mathbb{R}^3$ at perturbation parameter $\varepsilon = 1/1000$ whose orthogonal projections $K|S$ and $L|S$ onto every two-dimensional subspace $S \in G(3,2)$ are congruent (allowing planar reflections in $O(2)$), while K is neither a translate of L nor a translate of $-L$.

1. Construction and Verification

Following the two-cap support perturbation mechanism of Zhang (2017), let $B^2 \subset S^2$ be two disjoint spherical caps centered at non-antipodal, orthogonal unit vectors $u_1 = e_1$ and $u_2 = e_2$, and perturb the unit ball support function $h_{B^3}(u) = 1$ on the two caps and their antipodes with opposite parity signs at amplitude $\varepsilon = 1/1000$.

Theorem 1.1. *At $\varepsilon = 1/1000$, the support functions $h_K, h_L : S^2 \to \mathbb{R}$ satisfy:*

(1) Strict convexity: *the spherical Hessian $h + \nabla^2_{S^2} h$ is bounded below by $111/8000 > 0$ on S^2, so K and L are compact convex bodies with nonempty interior.*

(2) Universal planar projection congruence: *for every unit normal $n \in S^2$, the great circle $S_n = n^\perp \cap S^2$ intersects at most one perturbation cap pair at a time, and an explicit orthogonal transformation in $O(S_n)$ carries $h_K|_{S_n}$ to $h_L|_{S_n}$.*

(3) Exclusion of translations of $\pm L$: *evaluating $h_K - h_{\pm L}$ along the three spanning directions e_1, e_2, e_3 and their negatives forces any candidate translation vector $t \in \mathbb{R}^3$ to vanish, leaving an exact support gap of $1/8000 > 0$.*

Date: September 5, 2026.

2020 *Mathematics Subject Classification.* Primary 52A20; Secondary 52A15.